KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY

MICROPROCESSORS PROJECT REPORT

SMART MEDICINE REMINDER PROJECT

CODE

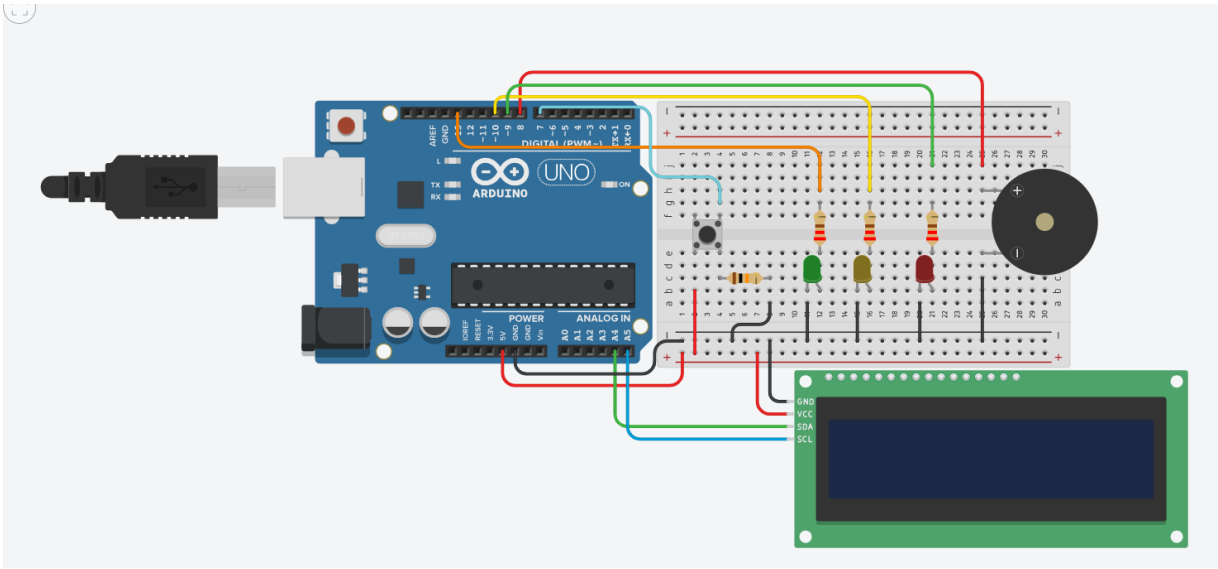
```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x27, 16, 2);

// Time between reminders after button press
const unsigned long WAIT_TIME = 5; // seconds

// Pins
const int buzzerPin = 8;
const int ledMorning = 13;
const int ledAfternoon = 10;
const int ledNight = 9;
const int buttonPin = 7;

// States
// 0 = wait for morning
// 1 = morning alarm, wait for button
// 2 = wait for afternoon
// 3 = afternoon alarm, wait for button
// 4 = wait for night
// 5 = night alarm, wait for button
// 6 = all done
int state = 0;

unsigned long stateStartTime = 0;
unsigned long lastBlinkTime = 0;
unsigned long lastDisplayTime = 0;

bool blinkState = false;
bool lastButtonState = LOW;

void setup() {
  lcd.init();
  lcd.backlight();

  pinMode(buzzerPin, OUTPUT);
  pinMode(ledMorning, OUTPUT);
  pinMode(ledAfternoon, OUTPUT);
  pinMode(ledNight, OUTPUT);
  pinMode(buttonPin, INPUT_PULLUP);
```

```
  digitalWrite(ledMorning, LOW);
  digitalWrite(ledAfternoon, LOW);
  digitalWrite(ledNight, LOW);

  Serial.begin(9600);

  lcd.setCursor(0, 0);
  lcd.print("SMART MEDICINE");
  lcd.setCursor(0, 1);
  lcd.print("REMINDER");
  delay(2000);
  lcd.clear();

  state = 0;
  stateStartTime = millis();
}

// Detect button press one time only
bool buttonPressed() {
  bool currentButtonState = digitalRead(buttonPin);

  if (lastButtonState == HIGH && currentButtonState == LOW) {
    delay(50); // debounce
    lastButtonState = currentButtonState;
    return true;
  }

  lastButtonState = currentButtonState;
  return false;
}

void loop() {
  unsigned long now = millis();
  unsigned long elapsedSeconds = (now - stateStartTime) / 1000;

  switch (state) {

    // -------------------------
    // WAIT FOR MORNING
    // -------------------------
    case 0:
      showCountdown("Next: Morning", WAIT_TIME - elapsedSeconds);
      if (elapsedSeconds >= WAIT_TIME) {
        state = 1;
        lcd.clear();
```

```
    resetAlarm();
    Serial.println("Morning reminder triggered");
  }
  break;

// -------------------------
// MORNING ALARM
// -------------------------
case 1:
  showAlarm("Take Morning Med");
  blinkLedAndBuzz(ledMorning);

  if (buttonPressed()) {
    noTone(buzzerPin);
    digitalWrite(ledMorning, HIGH);
    showTaken("Morning Taken");
    state = 2;
    stateStartTime = millis();   // next timer starts NOW
    lcd.clear();
    Serial.println("Morning medicine acknowledged");
  }
  break;

// -------------------------
// WAIT FOR AFTERNOON
// -------------------------
case 2:
  elapsedSeconds = (now - stateStartTime) / 1000;
  showCountdown("Next: Afternoon", WAIT_TIME - elapsedSeconds);
  if (elapsedSeconds >= WAIT_TIME) {
    state = 3;
    lcd.clear();
    resetAlarm();
    Serial.println("Afternoon reminder triggered");
  }
  break;

// -------------------------
// AFTERNOON ALARM
// -------------------------
case 3:
  showAlarm("Take After Med");
  blinkLedAndBuzz(ledAfternoon);

  if (buttonPressed()) {
    noTone(buzzerPin);
```

```
    digitalWrite(ledAfternoon, HIGH);
    showTaken("After Taken");
    state = 4;
    stateStartTime = millis();   // next timer starts NOW
    lcd.clear();
    Serial.println("Afternoon medicine acknowledged");
  }
  break;

// -------------------------
// WAIT FOR NIGHT
// -------------------------
case 4:
  elapsedSeconds = (now - stateStartTime) / 1000;
  showCountdown("Next: Night", WAIT_TIME - elapsedSeconds);
  if (elapsedSeconds >= WAIT_TIME) {
    state = 5;
    lcd.clear();
    resetAlarm();
    Serial.println("Night reminder triggered");
  }
  break;

// -------------------------
// NIGHT ALARM
// -------------------------
case 5:
  showAlarm("Take Night Med");
  blinkLedAndBuzz(ledNight);

  if (buttonPressed()) {
    noTone(buzzerPin);
    digitalWrite(ledNight, HIGH);
    showTaken("Night Taken");
    state = 6;
    lcd.clear();
    Serial.println("Night medicine acknowledged");
  }
  break;

// -------------------------
// ALL DONE
// -------------------------
case 6:
  noTone(buzzerPin);
  lcd.setCursor(0, 0);
```

```
    lcd.print(" All Medicines ");
    lcd.setCursor(0, 1);
    lcd.print("    Taken!     ");
    break;
  }
}

void showCountdown(const char* msg, long remaining) {
  if (remaining < 0) remaining = 0;

  if (millis() - lastDisplayTime >= 300) {
    lastDisplayTime = millis();

    lcd.setCursor(0, 0);
    lcd.print(msg);
    for (int i = strlen(msg); i < 16; i++) lcd.print(" ");

    lcd.setCursor(0, 1);
    lcd.print("Starts in: ");
    lcd.print(remaining);
    lcd.print("s   ");
  }
}

void showAlarm(const char* msg) {
  if (millis() - lastDisplayTime >= 300) {
    lastDisplayTime = millis();

    lcd.setCursor(0, 0);
    lcd.print("!! TAKE MED !! ");
    lcd.setCursor(0, 1);
    lcd.print(msg);
    for (int i = strlen(msg); i < 16; i++) lcd.print(" ");
  }
}

void blinkLedAndBuzz(int ledPin) {
  if (millis() - lastBlinkTime >= 300) {
    lastBlinkTime = millis();
    blinkState = !blinkState;

    digitalWrite(ledPin, blinkState);

    if (blinkState) {
      tone(buzzerPin, 1000);
    } else {
```

```
    noTone(buzzerPin);
   }
  }
}

void showTaken(const char* msg) {
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print(msg);
  lcd.setCursor(0, 1);
  lcd.print("Acknowledged");
  delay(1500);
}

void resetAlarm() {
  blinkState = false;
  lastBlinkTime = millis();
}
```

SCHEMATIC DIAGRAM OF THE PROJECT FROM TINKERCAD